

Reviewer 1

Comments

Reviewer 1 - Initial submission, 30/08/2026

I have reviewed the manuscript JMRT-D-26-07130 entitled “Influence of Varied Electrode Geometries on Machining Performance and Surface Integrity in Die-Sinking EDM of Al2219 Alloy”. I recommend Major Revision. My comments are as follows:

Introduction:

1. this part is not designed well.
2. Talk about general ideas in the field in the 1st paragraph and also in 1st paragraph you must provide more than one reference.
3. each paragraph should address a specific item. The paragraph design are not well enough.
4. also talk about your novelty and approach to the problem in the last paragraph of the study.
5. Table 1: it is a Figure not table label it as Fig. 1. Also, it seems the Figure is developed by AI. If yes mention it and the version.

Materials & Methods:

1. Provide references for all equations.
2. provide a graphical schematic of the experimental procedure.

Results:

1. XRD, and also microhardness test are recommended.
2. no compositional, phase, or elemental analysis (such as EDX or XRD) is provided to confirm the unverified carbon deposition mechanism phenomenon or to quantify the extent of carbon diffusion into the substrate surface.

Conclusion:

1. write your conclusion in a single paragraph about 200 words.

Other parts :

1. references, Author contributions, statements and other parts don't need number indexing. Remove the numbers.

Reviewer 1 - Revision 1, 03/10/2026

Major Revision

The authors have addressed most of my comments. However, the responses to the main scientific concerns remain incomplete. The following issues should be resolved before the manuscript can be considered for acceptance:

1. XRD analysis has not been provided.
2. The added EDX results support the presence of carbon on the analyzed surfaces but do not establish carbon diffusion into the substrate.

Reviewer 2

Comments

Reviewer 2 - Initial submission, 08/09/2026

The manuscript addresses a relevant die-sinking EDM problem for Al2219 blind holes and includes useful comparisons of MRR, TWR, machining time, overcut, Ra/Sa, OM/SEM/EDS, and 3D topography. The work has some practical values. However, the current version is not yet reliable enough for acceptance. A major revision is needed. The comments are as follows:

1. The same peak values are assigned to different electrodes in the abstract, Results, Discussion, and Conclusions e.g., DNC-4 vs DNC-6; groove width vs DNC number; “seven designs” vs nine electrodes/27 runs vs 21 samples. This undermines the core claims.
2. Table numbering and content are mismatched; Table 7 has missing/misaligned rows; the results table is corrupted; “surface micrograph” table is empty; Fig. 13 shows 21 samples while 27 experiments are claimed. Provide a clean experiment matrix and raw-data table.
3. Results appear single-run. Claims such as “significant,” “best,” and “164.6% improvement” require $n \geq 3$, mean $\pm$ SD/CI, ANOVA or equivalent tests, and multiple-comparison correction.
4. Electrodes have very different masses and active face areas but use the same 3/5/7 A. MRR differences may reflect current density/thermal mass rather than flushing geometry alone. Add constant-current-density or constant-face-area controls and report polarity, servo settings, flushing flow/temperature, endpoint control, alignment, and depth verification.
5. Surface integrity is underdeveloped relative to the title. Add white layer thickness, HAZ, cracks, recast layer, microhardness, and quantitative EDS mapping with calibration. Otherwise retitle toward “performance and surface topography.”
6. There exist many format errors in the manuscript. This should be wholly checked.

Reviewer 2 - Revision 1, 03/10/2026

The comments have been addressed.

Reviewer 3

Comments

Reviewer 3 - Initial submission, 03/09/2026

****Recommendation: Reject****

The manuscript presents an interesting experimental comparison of electrode geometries for EDM of Al2219; however, the novelty and fundamental scientific contribution are limited. The reported improvements in MRR and other machining responses are mainly process-oriented, without sufficient mechanistic understanding of discharge behavior, debris evacuation, material removal, or microstructural/surface evolution. The work therefore does not provide a sufficiently significant advancement in scientific understanding. I recommend rejection in its present form.